

Figure 7.5 Our map with our eight world cities added to it. At this distance, you can't tell how inaccurate these points are, but if you zoom in, you see that both of our Italian cities are actually in the Mediterranean.

7.1.2 Drawing points on a map

Projection isn't used only to display areas; it's also used to place individual points. Typically, you think of cities or people as represented not by their spatial footprint (though you do this with particularly large cities) but with a single point on a map, which is sized based on some variable such as population. A D3 projection can be used not only in a `geo.path()` but also as a function on its own. When you pass it an array with a pair of latitude and longitude coordinates, it returns the screen coordinates necessary to place that point. For instance, if we want to know where to place a point representing San Francisco (roughly speaking, -122 latitude, 37 longitude), then we could simply pass those values to our projection:

```
aProjection([-122,37])
```

[79.65586500535346, 194.32096033997914]

We can use this to add cities to our map along with loading the data from cities.csv, as in the following listing and which you see in figure 7.5.

Listing 7.6 Loading point and polygon geodata

```
queue()
    .defer(d3.json, "world.geojson")
    .defer(d3.csv, "cities.csv")
    .await(function(error, file1, file2) {
              createMap(file1, file2);
    });
```